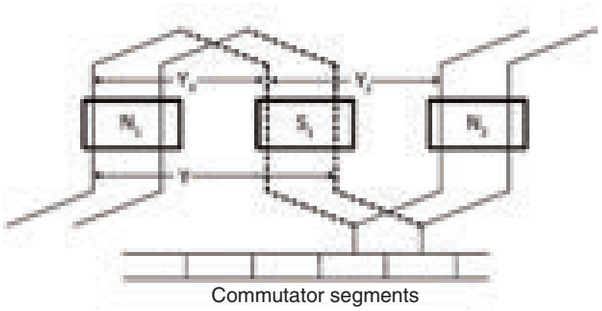


Fig. 9.6 **Wave winding**

9.5.5 Simplex lap winding

In simplex lap winding, finishing end of the first coil is connected with the starting end of the adjacent coil as shown in the figure 9.7.

Here, only one circuit is available.

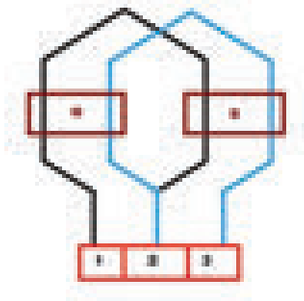


Fig. 9.7 **Simplex lap winding**

9.5.6 Duplex lap winding

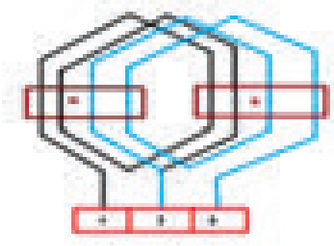


Fig. 9.8 **Duplex lap winding**

In duplex lap winding, finishing end of the first coil is connected with the starting end of the third coil as shown in the figure 9.8. There are two separated circuits in duplex windings.

Coils for the 1st circuit - 1,3,5,7, ...
Coils for the 2nd circuits - 2,4,6,8, ...

9.5.7 Back pitch (Y_b)

The distance between two coil sides of a coil is referred to as back pitch. It is always an odd number.

9.5.8 Winding pitch (or) coil pitch (y)

The distance between two starting leads of successive coils is referred to as winding pitch or coil pitch. It should be always an even number.

9.5.9 Front pitch (y_f)

It is the distance between two coil sides connected to the same commutator segment. It should be an odd number.

9.5.10 Necessary details needed for DC winding

i. First calculate the number of coils. The number of coils must be equal to the number of commutator segments.
ii. According to the formula concerned to the lap or wave winding, calculate the Back pitch (Y_b) Coil pitch (Y), and the Front pitch (Y_f)
iii. Back pitch and front pitch must be of odd number only and the coil pitch must be of even number.
iv. Formula for lap winding

$$\text{Back pitch, } Y_b = \frac{2C}{P} \pm K$$

Here, C – number of coils
P – number of poles

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